

A/B Testing - Complete Study Notes

1. What is A/B Testing?

A/B Testing (Split Testing) is a statistical experiment used to compare two versions (A and B) of a webpage, app, advertisement, or product feature to determine which performs better.

2. Objective

Make data-driven decisions by comparing two versions based on user behavior.

3. Control vs Treatment

Control (Version A): Original version.

Treatment (Version B): New or modified version.

4. Steps in A/B Testing

- Define the objective.
- Formulate hypotheses.
- Select the evaluation metric.
- Identify the population.
- Randomly assign users (Randomization).
- Decide the sample size.
- Run the experiment.
- Analyze results and draw conclusions.

5. Hypotheses

H0 (Null Hypothesis): No significant difference between A and B.

H1 (Alternative Hypothesis): A significant difference exists between A and B.

6. Metrics

Common metrics include Click-Through Rate (CTR), Conversion Rate, Revenue, Bounce Rate, and Time on Site.

7. Randomization

Randomly assigning participants removes selection bias and ensures a fair comparison.

8. Sample Size

A sufficiently large sample size improves the reliability of results and statistical power.

9. Statistical Decision

If $p\text{-value} < \alpha$ (usually 0.05), reject H_0 .

If $p\text{-value} \geq \alpha$, fail to reject H_0 .

10. Statistical Power

$\text{Power} = 1 - \beta$.

Higher power means a greater chance of detecting a true effect.

11. Type I and Type II Errors

Type I Error: False Positive.

Type II Error: False Negative.

12. Best Practices

- Test one major change at a time.
- Use random assignment.
- Run the experiment long enough.
- Ensure adequate sample size.
- Keep external conditions consistent.

13. Common Interview Questions

- What is A/B Testing?
- Difference between Control and Treatment?
- Why is randomization important?
- What does p-value represent?
- What is statistical power?

14. Acronyms

A/B = Version A and Version B

H0 = Null Hypothesis

H1 = Alternative Hypothesis

CTR = Click-Through Rate

CI = Confidence Interval

α = Significance Level

β = Probability of Type II Error

15. Quick Revision

A = Control

B = Treatment

Randomize users

Choose the correct metric

Use adequate sample size

$p < 0.05 \rightarrow$ Reject H0

Power = $1 - \beta$